



Hades

A Generalist Vision-Language-Action Model for Long-Horizon Real-Time GUI Game Agents

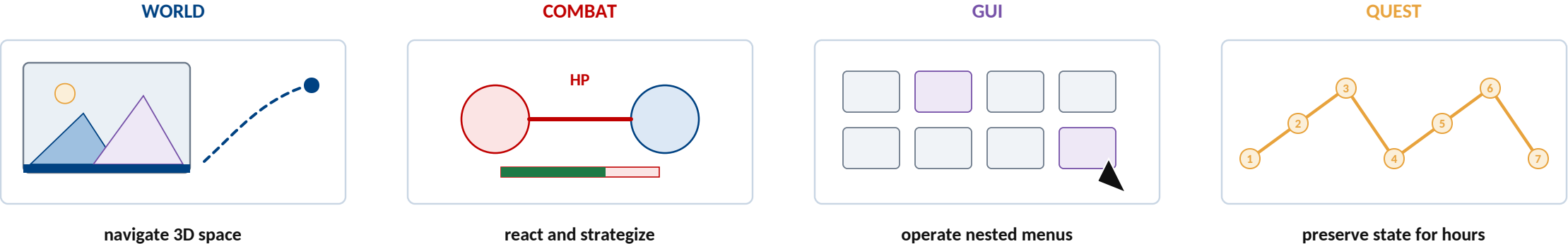


miHoYo Lumi · full author list in the paper's Contributions section

Industrial-scale generalist agent research

Manuscript dated 26 August 2026

Commercial games expose the full agent problem



One task couples four systems

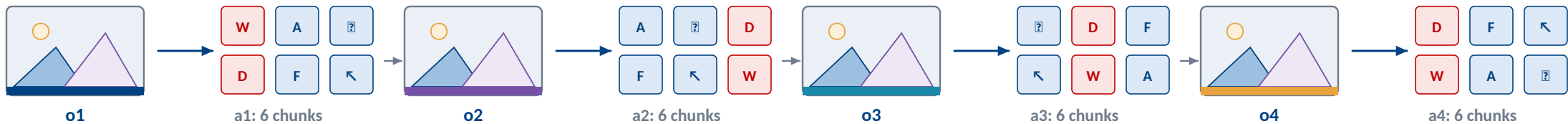


Pixels in, keyboard and mouse out — with **hours of state** in between.

Stage 1 — learn a universal device-level action language

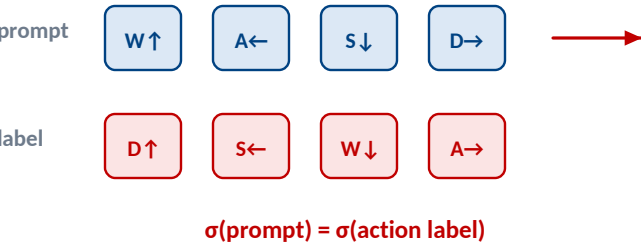
Raw interaction trajectory

visual 5 Hz • action 30 Hz

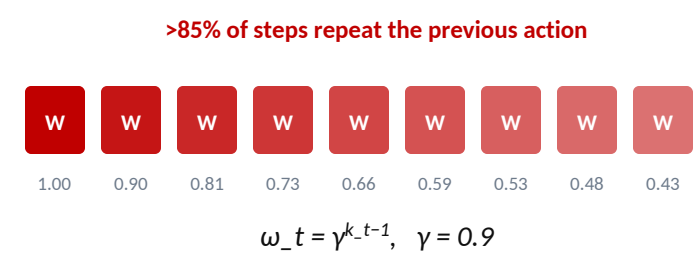


serialize <action_start> ΔX ΔY ΔZ ; K1 ; K2 ; K3 ; K4 ; K5 ; K6 <action_end>

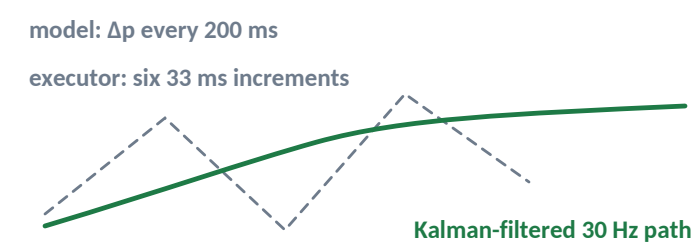
A Key Shuffle



B Repeated-action loss



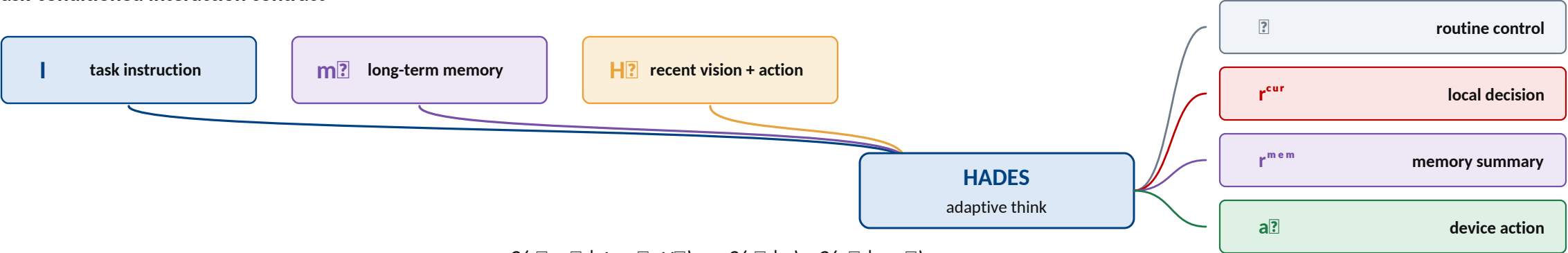
C Mouse inversion



Compress 30 Hz control into tokens, then debias what the model learns from repetition.

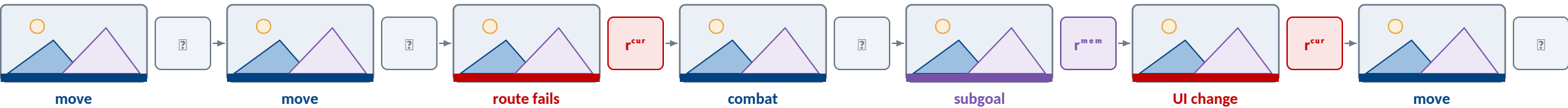
Stage 2 — think sparsely and remember across context windows

Task-conditioned interaction contract



$$\pi\theta(r^?, a^? | I, m^?, H^?) = \pi\theta(r^? | \cdot) \pi\theta(a^? | \cdot, r^?)$$

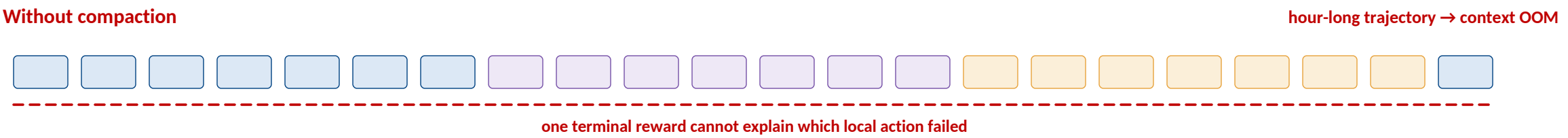
Sparse annotation over a continuous trajectory



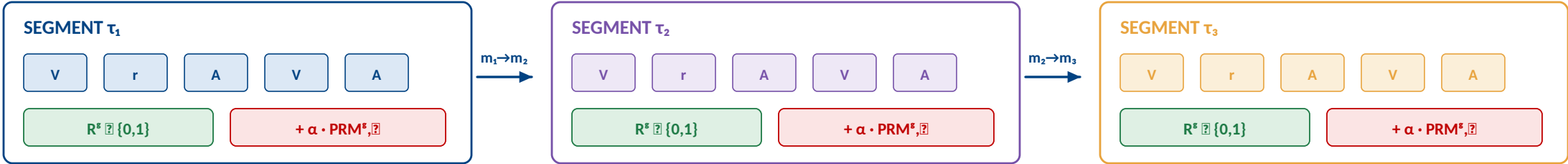
short-term queue keeps local detail • r^{mem} compacts completed stages into long-term memory

Routine control stays fast; decisions and memory appear only at **critical moments**.

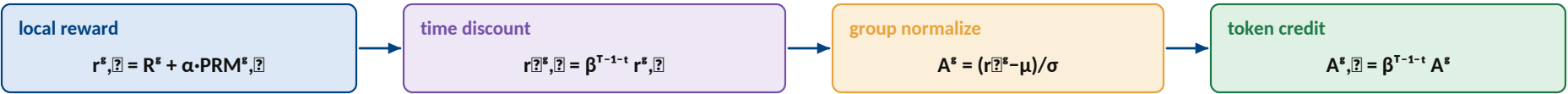
Stage 3 — TA-GRPO turns an hour-long rollout into learnable segments



TA-GRPO with memory-compacted segments



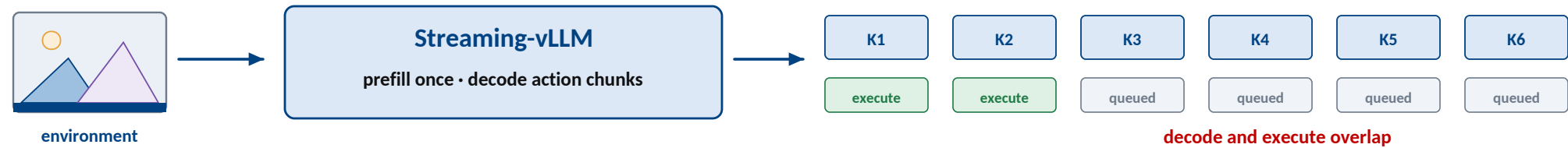
Trajectory-aware credit assignment



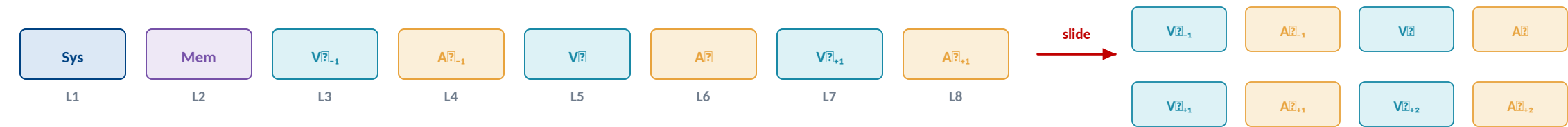
Compact memory across segments and assign credit near the state changes that matter.

Deployment — stream actions while the model is still decoding

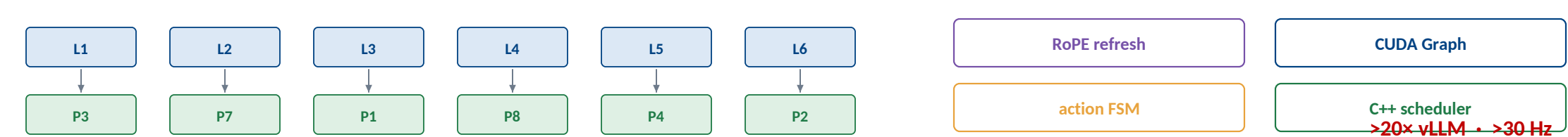
A Streaming client-server loop



B Sliding logical context

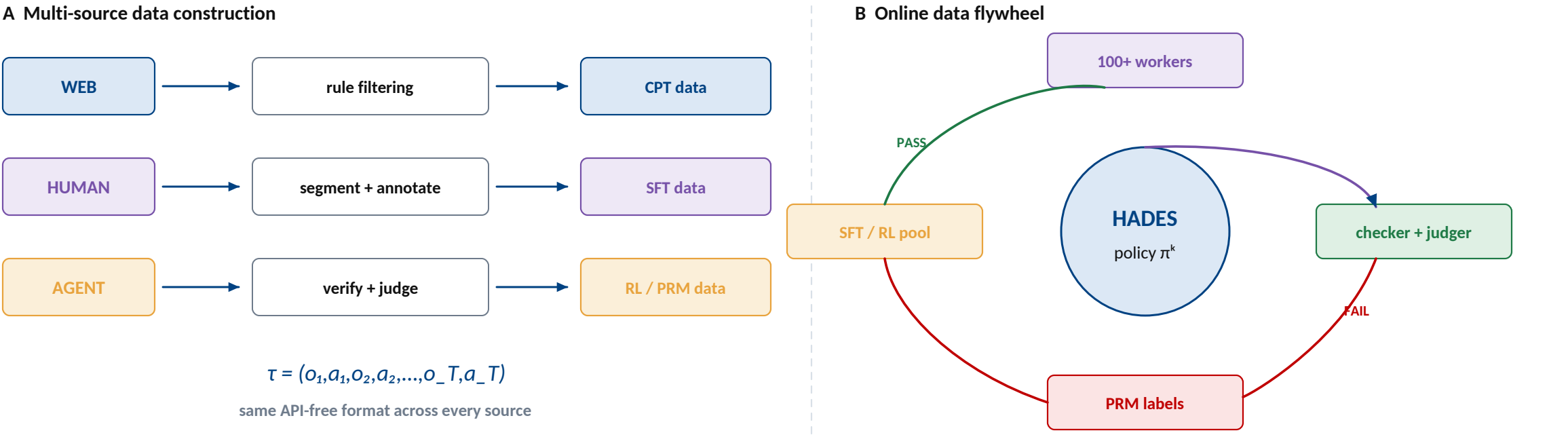


C Logical-to-physical KV lookup



Streaming-vLLM overlaps generation with execution and reuses the moving context.

System view — data, policy, verifier, and infrastructure form one loop



EVIDENCE BOUNDARY system metrics are reported; claimed benchmark leadership is not tabulated in the supplied manuscript

The contribution is an industrial agent stack, not only a policy model.